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Inventor(s)	Kawamura; Toshiaki et al.

Vehicle interior monitoring apparatus, vehicle interior monitoring method, and non-transitory computer-readable medium

Abstract

A vehicle interior monitoring apparatus (**10**) includes an acquisition unit (**110**), a detection unit (**120**), and a notification processing unit (**130**). The acquisition unit (**110**) repeatedly acquires, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle. The detection unit (**120**) performs, by using the sensor data, detection processing of a living body existing in the interior, each time the acquisition unit (**110**) acquires the sensor data. The notification processing unit (**130**) performs monitoring on an environment of the interior when a living body existing in the interior is detected, and performs notification for a pre-registered terminal (**40**) when the environment meets a first criterion.

Inventors:	Kawamura; Toshiaki (Tokyo, JP), Watanabe; Yoshihiro (Tokyo, JP), Suzuki; Yuichi (Tokyo, JP)
Applicant:	NEC Corporation (Tokyo, JP)
Family ID:	84322933
Assignee:	NEC CORPORATION (Tokyo, JP)
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Primary Examiner: Previl; Daniel

Background/Summary

(1) This application is a National Stage Entry of PCT/JP2021/021327 filed on Jun. 4, 2021, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a vehicle interior monitoring apparatus, a vehicle interior monitoring method, and a program.

BACKGROUND ART

(3) An environment (for example, temperature) inside a stopped vehicle tends to be harsh for a living body such as a person, a pet, and the like. Therefore, it is necessary to monitor an environment inside a vehicle when a living body remains inside the vehicle after the vehicle is stopped. For example, Patent Document 1 describes the following human detection apparatus. First, after a vehicle is stopped, the human detection apparatus estimates, by using a change rate of temperature inside the vehicle, a time in which it takes for the temperature inside the vehicle to reach a critical temperature. Then, the human detection apparatus outputs a human detection notice to an outside of the vehicle when a person is detected before reaching the time. In Patent Document 1, a mass sensor, a CCD camera, and a pyroelectric infrared sensor are exemplified as a human sensor.

RELATED DOCUMENT

Patent Document

(4) Patent Document 1: Japanese Patent Application Publication No. 2002-63668

SUMMARY OF INVENTION

Technical Problem

(5) Since a living body moves, the living body may be in a state in which the living body is undetected by a sensor even though the living body stays inside a vehicle after being detected by the sensor. One example of an object of the present invention is to output predetermined information to an outside of the vehicle when an environment inside the vehicle meets a criterion, even when the living body in the vehicle is in such a state.

Solution to Problem

(6) According to one aspect of the present invention, a vehicle interior monitoring apparatus is provided, including: an acquisition unit that repeatedly acquires, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; a detection unit that performs, by using the sensor data, detection processing of a living body existing in the interior, each time the acquisition unit acquires the sensor data; and a notification processing unit that performs monitoring on an environment of the interior when a living body existing in the interior is detected, and performs predetermined notification for a pre-registered terminal when the environment meets a first criterion.

(7) According to one aspect of the present invention, a vehicle interior monitoring method executed by a computer is provided. The vehicle interior monitoring method includes: acquisition processing of repeatedly acquiring, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; detection processing of performing, by using the sensor data, detection processing of a living body existing in the interior, each time the sensor data are acquired; and notification processing of performing monitoring on an environment of the interior when a living body existing in the interior is detected, and performing predetermined notification for a pre-registered terminal when the environment meets a first criterion.

(8) According to one aspect of the present invention, a program is provided, causing functions to be included, the functions are: an acquisition function of repeatedly acquiring, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; a

detection function of performing, by using the sensor data, detection processing of a living body existing in the interior, each time the acquisition function acquires the sensor data; and a notification processing function of performs monitoring on an environment of the interior when a living body existing in the interior is detected, and performing predetermined notification for a pre-registered terminal when the environment meets a first criterion.

Advantageous Effects of Invention

(9) According to the present invention, even when a living body in a vehicle is in a state in which the living body is undetected by a sensor after being detected by the sensor, predetermined information is output to an outside of the vehicle when an environment in the vehicle meets a criterion.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above-described object, another object, a feature and an advantage will be further clarified by suitable example embodiments described below and the following drawings accompanying thereto.

(2) FIG. 1 It is a diagram for describing a usage environment of a vehicle interior monitoring apparatus according to a first example embodiment.

(3) FIG. 2 It is a diagram illustrating one example of a functional configuration of the vehicle interior monitoring apparatus.

(4) FIG. 3 It is a diagram illustrating one example of a functional configuration of an onboard apparatus.

(5) FIG. 4 It is a diagram illustrating a hardware configuration example of the vehicle interior monitoring apparatus.

(6) FIG. 5 It is a diagram illustrating one example of processing performed by the onboard apparatus.

(7) FIG. 6 It is a diagram illustrating one example of processing performed by the vehicle interior monitoring apparatus.

(8) FIG. 7 It is a diagram for describing a usage environment of a vehicle interior monitoring apparatus according to a second example embodiment.

DESCRIPTION OF EMBODIMENTS

(9) In the following, example embodiments of the present invention will be described with reference to the drawings. Note that, in all the drawings, a similar component is denoted with a similar reference sign, and description thereof will not be repeated as appropriate.

First Example Embodiment

(10) FIG. 1 is a diagram for describing a usage environment of a vehicle interior monitoring apparatus **10** according to the present example embodiment. The vehicle interior monitoring apparatus **10** is used along with an onboard apparatus **20**. The vehicle interior monitoring apparatus **10** is, for example, a server installed outside a vehicle **30**.

(11) The onboard apparatus **20** repeatedly transmits, after the vehicle **30** is stopped and a door of the vehicle **30** is opened and then closed, data (hereinafter, referred to as sensor data) indicating a result of detection by a sensor of which a detection range is an interior of the vehicle **30** to the vehicle interior monitoring apparatus **10**. An interval of transmitting the sensor data is, for example, equal to or more than one second and equal to or less than one minute, but is not limited to this range.

(12) Each time acquiring sensor data from the onboard apparatus **20**, the vehicle interior monitoring apparatus **10** determines, by processing the sensor data, whether a living body, for example, a person such as a child or an elderly person, or an animal such as a pet remains in the

interior of the vehicle **30**. Then, when a living body remains, the vehicle interior monitoring apparatus **10** transmits information (hereinafter, referred to as alarm information) indicating that the living body remains to a terminal **40** when an environment of the interior of the vehicle **30** meets a predetermined condition.

(13) The terminal **40** is, for example, used by a user of the vehicle **30**. More specifically, the terminal **40** may be a terminal used by a parent of a child, or may be a terminal used by a child of an elderly person. Further the terminal **40** may be managed by a predetermined organization. The predetermined organization may be, for example, a public institution such as police or an organization entrusted by the police, or a private agency such as a security company.

(14) The vehicle **30** includes, as at least part of the sensor, an imaging apparatus **32**. The imaging apparatus **32** generates an image of the interior of the vehicle **30**. The onboard apparatus **20** transmits the image (hereinafter, referred to as an interior image) to the vehicle interior monitoring apparatus **10**, as part of sensor data. Note that, the onboard apparatus **20** may transmit, along with the interior image or instead of the interior image, feature data indicating a feature value of the interior image. Hereinafter, the interior image and/or the feature data is referred to as image data. Note that, a plurality of the imaging apparatuses **32** may be installed in the vehicle **30**. In this case, the plurality of imaging apparatuses **32** are installed at locations different from each another, and at least part of an imaging range differs from each other.

(15) Further, the vehicle **30** further includes a sensor **34**. The sensor **34** generates information (hereinafter, referred to as environmental information) indicating an environment of the interior of the vehicle **30**. One example of the environment is temperature. Then, the onboard apparatus **20** transmits, as part of the sensor data, the information generated by the sensor **34** to the vehicle interior monitoring apparatus **10**, in addition to an image generated by the imaging apparatus **32**.

(16) Further, the sensor **34** may include, for example, at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor. These sensors generate information (hereinafter, referred to as human detection data) indicating presence or absence of a living body existing in the interior of the vehicle **30**. The onboard apparatus **20** also transmits the human detection data to the vehicle interior monitoring apparatus **10**, as part of the sensor data.

(17) Further, the vehicle **30** includes an output apparatus **36**. The output apparatus **36** is, for example, a head lamp, a tail lamp, a brake lamp, a sound output apparatus (for example, a horn), and the like, and an output therefrom is recognizable for a person around the vehicle **30**. The output apparatus **36** may be a display visible from outside of the vehicle **30**. When transmitting alarm information to the terminal **40**, the vehicle interior monitoring apparatus **10** causes, as appropriate, via the onboard apparatus **20**, the output apparatus **36** to perform predetermined output. Note that, the output apparatus **36** may directly transmit alarm information to an external terminal (for example, the terminal **40**). In this case, the output apparatus **36** uses, for example, at least one of a telephone, an email, and SNS.

(18) Note that, the onboard apparatus **20** may communicate with the imaging apparatus **32**, the sensor **34**, and the output apparatus **36** directly, or communicate via a control apparatus installed in the vehicle **30**. Alternatively, the onboard apparatus **20** may be integrated with the imaging apparatus **32**.

(19) FIG. 2 is a diagram illustrating one example of a functional configuration of the vehicle interior monitoring apparatus **10**. The vehicle interior monitoring apparatus **10** includes an acquisition unit **110**, a detection unit **120**, and a notification processing unit **130**.

(20) The acquisition unit **110** acquires sensor data from the onboard apparatus **20**. As described above, the sensor data include at least image data and information generated by the sensor **34**.

(21) The detection unit **120** performs, by using the sensor data acquired by the acquisition unit **110**, detection processing of a living body existing in the interior of the vehicle **30**. For example, the detection unit **120** may perform the detection processing by using a model generated using machine learning, or may perform the detection processing by using a feature value of a living body (for

example, a feature value of a person, a feature value of a dog, and a feature value of a cat).

(22) The detection unit **120** may change a detection processing algorithm (or the model described above) for each type of living body. For example, when a living body is a person, the detection unit **120** may detect the living body by using an algorithm for face detection. In this case, the detection unit **120** may further estimate the age of the person.

(23) Further, the detection unit **120** may perform the detection processing using image data and the detection processing using information generated by the sensor **34** separately. Then, the detection unit **120** may determine that a living body exists in the interior of the vehicle **30** when the living body is detected in either one of the detection, or may determine that a living body exists in the interior of the vehicle **30** when the living body is detected in both of the detection.

(24) When a living body existing in the interior of the vehicle **30** is detected by the detection unit **120**, the notification processing unit **130** performs monitoring on an environment of the interior. The monitoring is performed by using environmental information included in sensor data. Then, when the environment of the interior of the vehicle **30** meets a first criterion, the notification processing unit **130** performs predetermined notification for the terminal **40**. The notification processing unit **130** preliminarily stores notification destination information (for example, an email address and SNS account information) for performing notification for the terminal **40**.

(25) Herein, the notification processing unit **130** may store notification destination information on a plurality of the terminals **40**, along with their priority. In this case, the notification processing unit **130** performs notification for a terminal **40** that has a highest priority. Then, when no reply is received from the notified terminal **40** within a predetermined time, the notification processing unit **130** performs notification for the terminal **40** that has a second highest priority. The notification processing unit **130** repeats the processing until a reply is received from the notified terminal **40** within the predetermined time.

(26) FIG. **3** is a diagram illustrating one example of a functional configuration of the onboard apparatus **20**. The onboard apparatus **20** includes a first communication unit **210** and a second communication unit **220**.

(27) The first communication unit **210** communicates with an apparatus inside the vehicle **30**. For example, the first communication unit **210** acquires, from the imaging apparatus **32**, an image generated by the imaging apparatus **32**, and also acquires, from the sensor **34**, data indicating a result of detection by the sensor **34**. Further, the first communication unit **210** outputs a signal for causing the output apparatus **36** to perform the predetermined output. A destination of the signal may be the output apparatus **36**, or may be a control apparatus of the output apparatus **36**. Note that, the first communication unit **210**, and the imaging apparatus **32** and the sensor **34** communicate with each other, for example, by using wireless, however, they may communicate with each other by using wire. Further, the first communication unit **210** and the output apparatus **36** (or the control apparatus of the output apparatus **36**) communicate with each other, for example, by using wire, however, they may communicate with each other by using wireless.

(28) Further, the first communication unit **210** acquires information indicating a state of the vehicle **30** from a control apparatus inside the vehicle **30**. For example, the first communication unit **210** acquires information indicating whether a power source of the vehicle **30** is stopped (for example, information indicating whether an engine or a motor is turned on or turned off) and information indicating that a door of the vehicle **30** is opened and closed.

(29) The second communication unit **220** communicates with the vehicle interior monitoring apparatus **10**. For example, the second communication unit **220** transmits sensor data to the vehicle interior monitoring apparatus **10**. Herein, the second communication unit **220** may generate feature data as necessary, by processing an image generated by the imaging apparatus **32**. Further, the second communication unit **220** may generate at least one of environmental information and human detection data as necessary, by processing data generated by the sensor **34**. Note that, these pieces of data may be the data itself generated by the sensor **34**.

(30) Further, the second communication unit **220** acquires, from the vehicle interior monitoring apparatus **10**, information for causing the output apparatus **36** to perform the predetermined output. The information is output to the output apparatus **36** (or the control apparatus of the output apparatus **36**) via the first communication unit **210**.

(31) FIG. **4** is a diagram illustrating a hardware configuration example of the vehicle interior monitoring apparatus **10**. The vehicle interior monitoring apparatus **10** includes a bus **1010**, a processor **1020**, a memory **1030**, a storage device **1040**, an input/output interface **1050**, and a network interface **1060**.

(32) The bus **1010** is a data transmission path for the processor **1020**, the memory **1030**, the storage device **1040**, the input/output interface **1050**, and the network interface **1060** to transmit and receive data to and from one another. However, a method of connecting the processor **1020** and the like with each other is not limited to bus connection.

(33) The processor **1020** is a processor achieved with a central processing unit (CPU), a graphics processing unit (GPU), and the like.

(34) The memory **1030** is a main storage apparatus achieved with a random access memory (RAM) and the like.

(35) The storage device **1040** is an auxiliary storage apparatus achieved with a hard disk drive (HDD), a solid state drive (SSD), a memory card, a read only memory (ROM), or the like. The storage device **1040** stores a program module that achieves each function of the vehicle interior monitoring apparatus **10** (for example, the acquisition unit **110**, the detection unit **120**, and the notification processing unit **130**). The processor **1020** reads each of the program modules onto the memory **1030** and executes the program module, and thereby each function related to the program module is achieved.

(36) The input/output interface **1050** is an interface for connecting the vehicle interior monitoring apparatus **10** and various pieces of input/output equipment with each other.

(37) The network interface **1060** is an interface for connecting the vehicle interior monitoring apparatus **10** to a network. The network is, for example, a local area network (LAN) or a wide area network (WAN). A method in which the network interface **1060** connects to the network may be wireless connection, or may be wired connection. The vehicle interior monitoring apparatus **10** communicates with the onboard apparatus **20** via the network interface **1060**.

(38) Note that, a hardware configuration of the onboard apparatus **20** is similar to the example illustrated in FIG. **5**. In this case, the storage device **1040** stores a program module that achieves the first communication unit **210** and the second communication unit **220**.

(39) FIG. **5** is a diagram illustrating one example of processing performed by the onboard apparatus **20**. The first communication unit **210** repeatedly acquires, from the control apparatus inside the vehicle **30**, information indicating a state of the vehicle **30**. When the second communication unit **220** detects, by using the information acquired by the first communication unit **210**, that the power source of the vehicle **30** is stopped, specifically, the vehicle **30** is stopped (step **S110**), the second communication unit **220** senses that a door of the vehicle **30** is opened then closed (step **S120**). Step **S120** indicates that a main occupant (for example, a driver) of the vehicle **30** has left the vehicle **30**. Then, the second communication unit **220** causes, via the first communication unit **210**, the imaging apparatus **32** to start imaging and the sensor **34** to start operating. Then, the second communication unit **220** starts transmitting sensor data to the vehicle interior monitoring apparatus **10** (step **S130**).

(40) Note that, as described above, the sensor data includes at least image data based on an image generated by the imaging apparatus **32**. Further, when the sensor **34** includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor, the sensor data also includes information generated by these sensors. The sensor data further includes environmental information, specifically, information indicating an environment (for example, temperature) of the interior of the vehicle **30**.

(41) Further, at least one of “the door of the vehicle **30** is locked” and “a predetermined time has elapsed since the door is no longer opened or closed” may further be added to a condition for the processing to transition from step **S120** to step **S130**.

(42) Further, “after a predetermined time has elapsed since a person in a driver's seat is no longer detected” may be used as a condition for the processing to proceed from step **S110** to step **S130**.

(43) Then, until a door is opened again, specifically, until a person returns to the vehicle **30** (step **S140**), the onboard apparatus **20** continues transmitting sensor data (step **S150**). In other words, the onboard apparatus **20** continues transmitting sensor data from when a person leaves the vehicle **30** until when the person returns.

(44) Note that, the onboard apparatus **20** may transmit only the environmental information to the vehicle interior monitoring apparatus **10** after the vehicle interior monitoring apparatus **10** detects a living body remaining in the interior of the vehicle **30**. To do so, after detecting the living body, the vehicle interior monitoring apparatus **10** may transmit information indicating that the living body is detected to the onboard apparatus **20**.

(45) FIG. **6** is a diagram illustrating one example of processing performed by the vehicle interior monitoring apparatus **10**. When the vehicle interior monitoring apparatus **10** starts receiving sensor data and environmental information from the onboard apparatus **20**, the vehicle interior monitoring apparatus **10** performs the processing illustrated in the present drawing. As described above, the onboard apparatus **20** continues to transmit sensor data from when a person leaves the vehicle **30** until the person returns. Thus, the vehicle interior monitoring apparatus **10** performs the processing illustrated in the present drawing from when a person leaves the vehicle **30** until the person returns.

(46) First, the acquisition unit **110** acquires sensor data from the onboard apparatus **20**. Then, the detection unit **120** acquires image data out of the sensor data (step **S210**), and determines, by processing the image data, presence or absence of a living body in the interior of the vehicle **30** (step **S220**). When no living body is detected from the piece of image data processed this time (step **S220**: No), the detection unit **120** performs similar processing for a piece of image data acquired next by the acquisition unit **110** (steps **S210**, **S220**).

(47) Note that, in steps **S210** and **S220**, human detection data generated by the sensor **34** may be used along with the image data.

(48) On the other hand, when the detection unit **120** detects a living body (step **S220**: Yes), the notification processing unit **130** acquires environmental information transmitted along with the piece of image data from which the living body is sensed (step **S230**), and determines, by processing the environmental information, whether an environment of the interior meets a first criterion (step **S240**). The first criterion may be, for example, that temperature of the interior exceeds a first reference value (for example, a value equal to or more than 30° C. and equal to or less than 40° C.), or that the temperature of the interior falls below a second reference value (for example, a value equal to or more than 0° C. and equal to or less than 10° C.) being set lower than the first reference value.

(49) When the environment of the interior does not meet the first criterion (step **S240**: No), the notification processing unit **130** performs a similar processing for a piece of environmental information acquired next from the onboard apparatus **20** by the acquisition unit **110** (steps **S230**, **S240**). At this occasion, the detection unit **120** may stop the detection processing of a living body until a door of the vehicle **30** is opened again. A reason for this is, for example, that a living body may be out of the imaging range of the imaging apparatus **32**, or that a living body may get under a blanket and the like. In this case, when the detection unit **120** continuously performs the detection processing of a living body, it may be incorrectly determined that a living body is not present in the interior, even though the living body is present in the interior.

(50) On the other hand, when the environment of the interior meets the first criterion (step **S240**: Yes), the notification processing unit **130** transmits information indicating that the first criterion is

met, specifically, alarm information, to the terminal **40** (step **S250**). After that, the notification processing unit **130** may repeatedly transmit the alarm information until a user of the terminal **40** confirms the alarm information (for example, until a predetermined reply is received from the terminal **40**). Then, the vehicle interior monitoring apparatus **10** may end the processing when the predetermined reply is received from the terminal **40**.

(51) Herein, when the notification processing unit **130** stores information on a plurality of the terminals **40**, the notification processing unit **130** performs notification for a terminal **40** that has a highest priority. Then, when no reply is received from the notified terminal **40** within a predetermined time, notification is performed for a terminal **40** that has a second highest priority. The notification processing unit **130** repeats the processing until a reply is received from the notified terminal **40** within a predetermined time.

(52) Further, the notification processing unit **130** may perform the processing described as the step **S250** when a state in which the first criterion is met continues for a predetermined time.

(53) Further, when a living body is a person, the detection unit **120** may estimate age of the detected person in step **S220**. Then, the notification processing unit **130** may further add “the age meets a second criterion” to a condition for transmitting the alarm information. The second criterion is, for example, that the person is 10 years old or younger, the person is 70 years old or older, or that either one of these two conditions is met.

(54) Further, in step **S250**, the notification processing unit **130** may further transmit, to the onboard apparatus **20**, information for causing the output apparatus **36** to perform the predetermined output. In this case, the onboard apparatus **20** causes the output apparatus **36** to perform the predetermined output. In this way, a person around the vehicle **30** can recognize that a living body exists in the vehicle **30**.

(55) Further, the detection unit **120** may determine, by processing image data when the environment of the interior meets the first criterion, whether a living body is currently captured by the imaging apparatus **32**. Then, the notification processing unit **130** may include a result of the determination in alarm information. In this way, a user of the terminal **40** can easily estimate where in the interior the living body is.

(56) Note that, in the processing illustrated in the FIG. 5, the onboard apparatus **20** transmits sensor data to the vehicle interior monitoring apparatus **10** after the power source is turned off and a door is opened and closed at least once. However, the onboard apparatus **20** may constantly transmit sensor data to the vehicle interior monitoring apparatus **10**. In this case, the onboard apparatus **20** transmits information indicating that the power source is turned on or turned off to the vehicle interior monitoring apparatus **10** at timing when the power source is turned on or turned off, and also transmits information indicating that a door is opened and closed to the vehicle interior monitoring apparatus **10** at timing when the door is opened and closed. The vehicle interior monitoring apparatus **10** selects, from the sensor data acquired from the onboard apparatus **20**, data after the power source is turned off and a door is opened and closed at least once, by using these pieces of information, and performs the processing illustrated in FIG. 6.

(57) As described above, according to the present example embodiment, the vehicle interior monitoring apparatus **10** first determines, by using sensor data, whether a living body is present in the interior of the vehicle **30**. Then, when a living body is present in the interior of the vehicle **30**, the vehicle interior monitoring apparatus **10** performs monitoring, by using environmental information, whether an environment of the interior of the vehicle **30** meets the first criterion. During the monitoring, the vehicle interior monitoring apparatus **10** does not use the sensor data. Specifically, once the vehicle interior monitoring apparatus **10** senses a living body, the vehicle interior monitoring apparatus **10** continues to monitor the environment in the interior, even when the living body is no longer sensed. Therefore, the vehicle interior monitoring apparatus **10** can transmit predetermined information to the terminal **40** when the environment in the vehicle meets the first criterion, even when a living body in the interior is in a state in which the living body is

not easily detected by the imaging apparatus **32** and the sensor **34**.

(58) Further, the vehicle interior monitoring apparatus **10** acquires sensor data after the power source of the vehicle **30** is turned off and a door is opened and closed at least once, and processes the sensor data. Therefore, a communication traffic volume between the vehicle interior monitoring apparatus **10** and the onboard apparatus **20** is reduced, and an amount of computation by the vehicle interior monitoring apparatus **10** is also reduced. Further, the vehicle interior monitoring apparatus **10** stops the detection processing of a living body after detecting a living body. Therefore, the amount of computation by the vehicle interior monitoring apparatus **10** is further reduced.

Second Example Embodiment

(59) FIG. **7** is a diagram for describing a usage environment of a vehicle interior monitoring apparatus **10** according to the present example embodiment. In the example illustrated in the present drawing, the vehicle interior monitoring apparatus **10** is installed inside a vehicle **30**. In other words, an onboard apparatus **20** illustrated in FIG. **1** includes a function of the vehicle interior monitoring apparatus **10**. Also in this way, an advantageous effect similar to that of the first example embodiment can be achieved.

(60) While the example embodiments of the present invention have been described with reference to the drawings, the example embodiments are examples of the present invention, and various configurations other than those described above may be adopted.

(61) Further, although, a plurality of steps (pieces of processing) are described in order in the plurality of flowcharts referred to in the above description, an order of executing the steps in each example embodiment is not limited to an order described in the flowcharts. In each example embodiment, an order of the illustrated steps may be changed to an extent that contents thereof are not interfered. Further, each of the above-described example embodiments can be combined to an extent that contents thereof do not conflict with each other.

(62) A part or the entirety of the above-described example embodiments may be described as the following supplementary notes, but is not limited thereto.

(63) 1. A vehicle interior monitoring apparatus including:

(64) an acquisition unit that repeatedly acquires, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; a detection unit that performs, by using the sensor data, detection processing of a living body existing in the interior, each time the acquisition unit acquires the sensor data; and a notification processing unit that performs monitoring on an environment of the interior when a living body existing in the interior is detected, and performs predetermined notification for a pre-registered terminal when the environment meets a first criterion.

2. The vehicle interior monitoring apparatus according to supplementary note 1, in which the sensor includes an imaging apparatus.

3. The vehicle interior monitoring apparatus according to supplementary note 2, in which the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.

4. The vehicle interior monitoring apparatus according to any one of supplementary notes 1 to 3, in which the detection unit stops the detection processing after a living body existing in the interior is detected until a door of the vehicle is opened.

5. The vehicle interior monitoring apparatus according to any one of supplementary notes 1 to 4, in which the environment includes temperature.

6. The vehicle interior monitoring apparatus according to any one of supplementary notes 1 to 5, in which, when the environment exceeds the first criterion, the notification processing unit further causes an output apparatus installed in the vehicle to perform predetermined output.

7. The vehicle interior monitoring apparatus according to any one of supplementary notes 1 to 6, in

which the living body includes at least one of a person and an animal.

8. The vehicle interior monitoring apparatus according to supplementary note 7, in which the detection unit estimates age of a person when the living body is the person, and the notification processing unit performs the notification when the age meets a second criterion.

9. A vehicle interior monitoring method including, executing by a computer: acquisition processing of repeatedly acquiring, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; detection processing of performing, by using the sensor data, detection on a living body existing in the interior, each time the sensor data are acquired; and notification processing of performing monitoring on an environment of the interior when a living body existing in the interior is detected, and performing predetermined notification for a pre-registered terminal when the environment meets a first criterion.

10. the vehicle interior monitoring method according to supplementary note 9, in which the sensor includes an imaging apparatus.

11. The vehicle interior monitoring method according to supplementary note 10, in which the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.

12. The vehicle interior monitoring method according to any one of supplementary notes 9 to 11, further including by the computer, stopping the detection processing after a living body existing in the interior is detected until a door of the vehicle is opened.

13. The vehicle interior monitoring method according to any one of supplementary notes 9 to 12, in which the environment includes temperature.

14. The vehicle interior monitoring method according to any one of supplementary notes 9 to 13, further including by the computer, in the notification processing, further causing an output apparatus installed in the vehicle to perform predetermined output when the environment exceeds the first criterion.

15. The vehicle interior monitoring method according to any one of supplementary notes 9 to 14, in which the living body includes at least one of a person and an animal.

16. The vehicle interior monitoring method according to supplementary note 15, further including, by the computer: in the detection processing, estimating age of a person when the living body is the person; and in the notification processing, performing the notification when the age meets a second criterion.

17. A program causing a computer to include: an acquisition function of repeatedly acquiring, after a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data indicating a result of detection by a sensor of which a detection range is an interior of the vehicle; a detection function of performing, by using the sensor data, detection processing of a living body existing in the interior, each time the acquisition function acquires the sensor data; and a notification processing function of performing monitoring on an environment of the interior when a living body exiting in the interior is detected, and performs predetermined notification for a pre-registered terminal when the environment meets a for a first criterion.

18. The program according to supplementary note 17, in which the sensor includes an imaging apparatus.

19. The program according to supplementary note 18, in which the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.

20. The program according to any one of supplementary notes 17 to 19, in which the detection function stops the detection processing after a living body existing in the interior is detected until a door of the vehicle is opened.

21. The program according to any one of supplementary notes 17 to 20, in which the environment includes temperature.

22. The program according to any one of supplementary notes 17 to 21, in which, when the

environment exceeds the first criterion, the notification processing function further causes an output apparatus installed in the vehicle to perform predetermined output.

23. The program according to any one of supplementary notes 17 to 22, in which the living body includes at least one of a person and an animal.

24. The program according to supplementary note 23, in which the detection function estimates age of a person when the living body is the person, and the notification processing function performs the notification when the age meets a second criterion.

REFERENCE SIGNS LIST

(65) **10** Vehicle interior monitoring apparatus **20** Onboard apparatus **30** Vehicle **32** Imaging apparatus **34** Sensor **36** Output apparatus **40** Terminal **110** Acquisition unit **120** Detection unit **130** Notification processing unit **210** First communication unit **220** Second communication unit

Claims

1. A vehicle interior monitoring apparatus comprising: at least one memory storing instructions; and at least one processor configured to execute the instructions to perform operations comprising: acquiring, in a case where a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data generated by a sensor monitoring an interior of the vehicle; performing, by using the sensor data, detection processing of a living body existing in the interior, each time the sensor data is acquired; notifying a terminal based on a result of the detection processing; and stopping, in a case where the living body is detected, the detection processing until the door is opened again.
2. The vehicle interior monitoring apparatus according to claim 1, wherein the sensor includes an imaging apparatus.
3. The vehicle interior monitoring apparatus according to claim 2, wherein the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.
4. The vehicle interior monitoring apparatus according to claim 1, wherein the operations further comprise: performing, by using the sensor data, monitoring an environment of the interior; and notifying the terminal in a case where the environment after the living body is detected meets a first criterion.
5. The vehicle interior monitoring apparatus according to claim 4, wherein the environment includes temperature.
6. The vehicle interior monitoring apparatus according to claim 4, wherein the operations further comprise: performing the notification in a case where the environment exceeds the first criterion.
7. The vehicle interior monitoring apparatus according to claim 1, wherein the operations further comprise: causing an output apparatus installed in the vehicle to perform output based on the result of the detection processing.
8. The vehicle interior monitoring apparatus according to claim 1, wherein the living body includes at least one of a person and an animal.
9. The vehicle interior monitoring apparatus according to claim 8, wherein the operations further comprise estimating an age of a person in a case where the living body is the person, and performing the notification in a case where the age meets a second criterion.
10. The vehicle interior monitoring apparatus according to claim 1, wherein the operations further comprise: acquiring the sensor data in a case where a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, and the door is locked.
11. The vehicle interior monitoring apparatus according to claim 1, wherein the operations further comprise: notifying a first terminal; receiving a reply from the first terminal; and notifying a second terminal in a case where no reply is received from the first terminal within a predetermined time.

12. A vehicle interior monitoring method executed performed by a computer and comprising: acquiring, in a case where a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data generated by a sensor monitoring an interior of the vehicle; performing, by using the sensor data, detection of a living body existing in the interior, each time the sensor data is acquired; notifying a terminal based on a result of the detection processing; and stopping, in a case where the living body is detected, the detection processing until the door is opened again.
 13. The vehicle interior monitoring method according to claim 12, wherein the sensor includes an imaging apparatus.
 14. The vehicle interior monitoring method according to claim 13, wherein the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.
 15. The vehicle interior monitoring method according to claim 12, further comprising: performing, by using the sensor data, monitoring an environment of the interior; and notifying the terminal in a case where the environment after the living body is detected meets a first criterion.
 16. The vehicle interior monitoring method according to claim 15, wherein the environment includes temperature.
 17. A non-transitory computer-readable medium storing a program executable by a computer to perform operations comprising: acquiring, in a case where a power source of a vehicle is turned off and a door of the vehicle is opened and closed at least once, sensor data generated by a sensor monitoring an interior of the vehicle; performing, by using the sensor data, detection processing of a living body existing in the interior, each time the sensor data is acquired; notifying a terminal based on a result of the detection processing; and stopping, in a case where the living body is detected, the detection processing until the door is opened again.
 18. The non-transitory computer-readable medium according to claim 17, wherein the sensor includes an imaging apparatus.
 19. The non-transitory computer-readable medium according to claim 18, wherein the sensor further includes at least one of a millimetric wave sensor, an infrared sensor, a temperature sensor, and a vibration sensor.
 20. The non-transitory computer-readable medium according to claim 17, wherein the operations further comprise: performing, by using the sensor data, monitoring an environment of the interior; and notifying the terminal in a case where the environment after the living body is detected meets a first criterion.
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